# A Mixed-Effect Scaling Model for Kinematic Slip-Fields

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Table S1: Standard deviation of random-effects for different scaling parameters

| **Parameter** | **Intercept** | | | | **Slope** | | | |
|---|---|---|---|---|---|---|---|---|
| | FM | SVT | TS | NM | FM | SVT | TS | NM |
| $L$ | 0.1360 | 0.2798 | – | – | 0.0190 | 0.0297 | – | – |
| $W$ | 0.5439 | 0.0042 | 0.1459 | 0.0650 | 0.0884 | 0.0059 | 0.0096 | 0.0156 |
| $A$ | 0.6931 | 0.2058 | 0.4603 | 0.4510 | 0.1031 | 0.0187 | 0.0447 | 0.0656 |
| $D_{\text{mean}}$ | 0.4580 | 0.1094 | 0.5162 | 0.3893 | 0.0761 | 0.0049 | 0.0501 | 0.0640 |
| $L_{\text{eff}}$ | 0.1807 | – | – | 0.5622 | 0.0312 | – | – | 0.0733 |
| $W_{\text{eff}}$ | 0.4395 | 0.1386 | 0.0635 | 0.1377 | 0.0740 | 0.0233 | 0.0144 | 0.0237 |
| $A_{\text{eff}}$ | 0.5710 | 0.0896 | 0.1167 | 0.4705 | 0.0886 | 0.0202 | 0.0070 | 0.0680 |
| $D_{\text{eff}}$ | 0.3807 | 0.1898 | 0.2409 | 0.5305 | 0.0638 | 0.0350 | 0.0212 | 0.0812 |
| $\sigma_D$ | 0.4637 | 0.1296 | 0.1680 | 0.4974 | 0.0740 | 0.0256 | 0.0174 | 0.0757 |
| $A_{\text{LA}}$ | 0.5752 | – | – | – | 0.0880 | – | – | – |
| $A_{\text{VLA}}$ | 0.6065 | 0.7893 | – | – | 0.0892 | 0.0917 | – | – |
| $A_{\text{CA}}$ | 0.7086 | – | – | 0.4689 | 0.1049 | – | – | 0.0661 |
| $R_{\text{CVLA}}$ | – | 1.0990 | – | – | – | 0.1388 | – | – |
| $R_{\text{CCA}}$ | – | 0.5745 | – | 0.8361 | – | 0.0687 | – | 0.1007 |
| $R_{\text{dmax}}$ | 0.1613 | – | – | – | 0.0105 | – | – | – |
| $R_{\text{LA}}$ | – | – | – | 0.7052 | – | – | – | 0.0814 |
| $R_{\text{VLA}}$ | – | 0.8944 | – | – | – | 0.1134 | – | – |
| $ax_{\text{exp}}$ | 0.5006 | 0.5057 | – | – | 0.0673 | 0.0572 | – | – |
| $ax_{\text{von}}$ | 0.1640 | 0.7304 | – | – | 0.0265 | 0.1070 | – | – |
| $ax_{\text{gauss}}$ | 0.3360 | 0.1439 | 0.3493 | 0.7006 | 0.0452 | 0.0149 | 0.0382 | 0.0896 |
| $az_{\text{exp}}$ | 0.6006 | 0.4017 | – | – | 0.0928 | 0.0402 | – | – |
| $az_{\text{von}}$ | 0.3638 | 1.1170 | – | – | 0.0573 | 0.1577 | – | – |
| $az_{\text{gauss}}$ | 0.2989 | 0.3663 | – | 0.5573 | 0.0403 | 0.0397 | – | 0.0698 |

Table S2: Comparison of RMSE and AIC values for various source scaling models across fault types and parameters. *Note: The best model for each parameter has been highlighted in bold.*

| **Model** | **SS Fault** | | **Thrust Fault** | | **Normal Fault** | |
|---|---|---|---|---|---|---|
| | **RMSE** | **AIC** | **RMSE** | **AIC** | **RMSE** | **AIC** |
| ***W (Rupture Width)*** | | | | | | |
| Mai and Beroza (2000) | 0.36 | −69.27 | 0.27 | −227.96 | 0.20 | −28.07 |
| Raghukanth and Sangeetha (2016) | 0.38 | −65.54 | 0.21 | −270.04 | 0.14 | **−35.61** |
| Wells and Coppersmith (1994) | 0.38 | −65.43 | 0.53 | −109.86 | 0.39 | −15.08 |
| This study | **0.23** | **−99.15** | **0.18** | **−302.17** | **0.13** | −34.67 |
| $L_{\text{eff}}$ ***(Effective Rupture Length)*** | | | | | | |
| Mai and Beroza (2000) | 0.61 | −48.12 | **0.49** | **−182.51** | 0.75 | −10.66 |
| Raghukanth and Sangeetha (2016) | **0.48** | **−74.12** | 0.51 | −169.18 | **0.55** | **−27.50** |
| Goda et al. (2016) | 0.53 | −63.03 | 0.53 | −157.82 | 0.62 | −20.65 |
| This study | 0.55 | −57.70 | 0.55 | −148.76 | 0.62 | −19.01 |
| $a_{x,\text{von}}$ ***(Along-strike Von-Karman correlation length)*** | | | | | | |
| Mai and Beroza (2000) | 0.33 | −80.30 | 0.18 | −352.25 | — | |
| Raghukanth and Sangeetha (2016) | 0.37 | −72.46 | 0.31 | −239.21 | — | |
| Goda et al. (2016) | 0.27 | **−94.23** | 0.19 | −344.62 | — | |
| This study | **0.27** | −94.19 | **0.18** | **−356.22** | — | |
| $a_{z,\text{von}}$ ***(Down-dip Von-Karman correlation length)*** | | | | | | |
| Mai and Beroza (2000) | 0.29 | −90.56 | 0.20 | −329.13 | — | |
| Raghukanth and Sangeetha (2016) | 0.35 | −74.79 | 0.43 | −172.67 | — | |
| Goda et al. (2016) | 0.25 | −102.35 | 0.18 | −354.65 | — | |
| This study | **0.24** | **−103.89** | **0.15** | **−388.15** | — | |
| $A_{\text{vla}}$ ***(Area of very large asperity)*** | | | | | | |
| Mai and Beroza (2000) | 0.33 | −79.40 | 0.52 | −132.30 | — | |
| Raghukanth and Sangeetha (2016) | 0.44 | −57.60 | 0.31 | −244.76 | — | |
| This study | **0.26** | **−97.09** | **0.28** | **−264.36** | — | |
| $W_{\text{eff}}$ ***(Effective Rupture Width)*** | | | | | | |
| Mai and Beroza (2000) | 0.44 | −55.06 | 0.49 | −122.92 | — | |
| Raghukanth and Sangeetha (2016) | 0.45 | −52.77 | 0.49 | −124.31 | — | |
| Goda et al. (2016) | **0.40** | **−61.64** | 0.53 | −110.67 | — | |
| This study | 0.42 | −55.71 | **0.46** | **−131.57** | — | |
| $A_{LA}$ ***(Area of large asperity)*** | | | | | | |
| Mai and Beroza (2000) | 0.27 | −239.62 | 0.39 | −411.08 | — | |
| Raghukanth and Sangeetha (2016) | 0.38 | −180.20 | 0.32 | −500.49 | — | |
| This study | **0.27** | **−241.71** | **0.30** | **−540.13** | — | |
| $R_{LA}$ ***(Distance to large asperity)*** | | | | | | |
| Raghukanth and Sangeetha (2016) | 0.44 | −492.81 | 0.41 | −410.41 | — | |
| This study | **0.26** | **−809.08** | **0.36** | **−470.26** | — | |

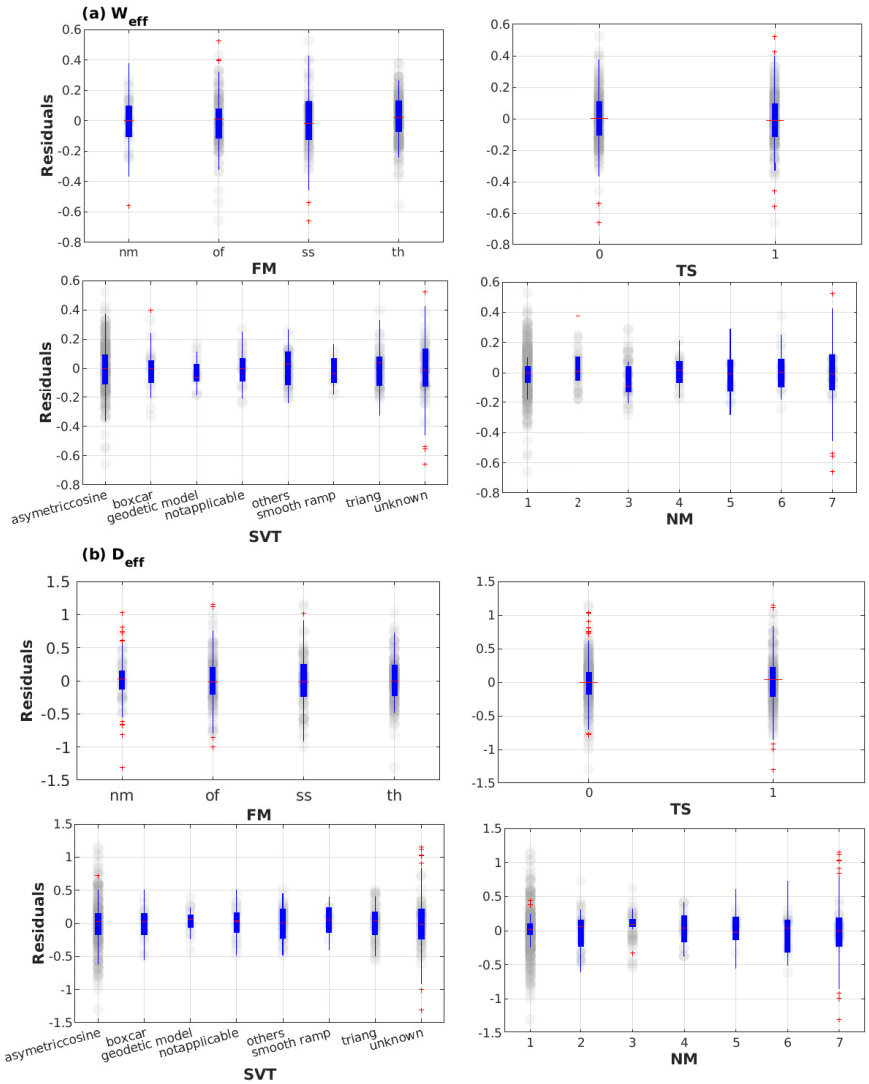


Figure S1: Box-plot showing the residual distribution for (a) $W_{eff}$ and (b) $D_{eff}$ along different random-effect levels. In the plot, the grey color represents the $log_{10}$ residuals population. Note: Please refer to Table 2 of the manuscript for the mixed-effect functional form of the scaling parameter.

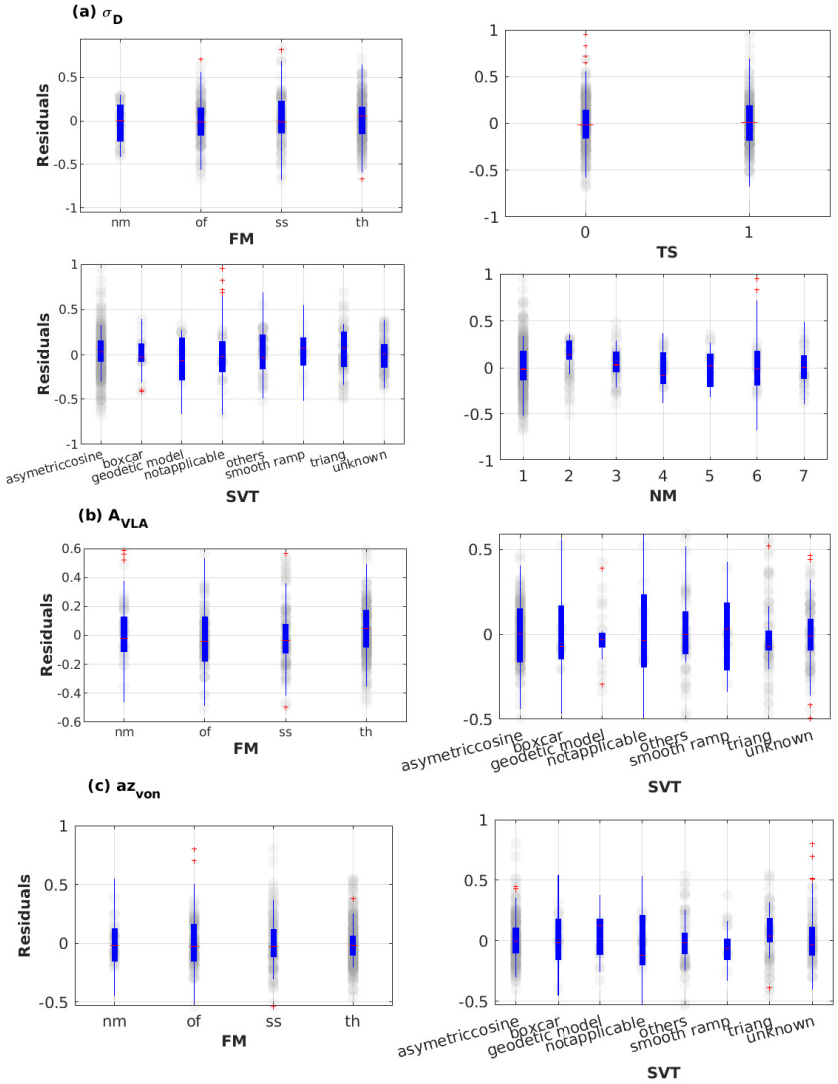


Figure S2: Box-plot showing the residual distribution for (a) $\sigma_D$, (b) $A_{LA}$ and (c) $az_{von}$ along different random-effect levels. In the plot, the grey color represents the $log_{10}$ residuals population. Note: Please refer to Table 2 of the manuscript for the mixed-effect functional form of the scaling parameter.

# 1 Stepwise Procedure for outlier diagnostics

A stepwise procedure for outlier-based diagnostics, statistical testing, and checking for sensitivity of the results is demonstrated for scaling parameter $W$, which is below:
Initially, the model is trained on the entire dataset, from which residuals are obtained and standardized. Outliers beyond the 2.5 threshold are removed. The residuals distribution for the two datasets is shown in the figure below:

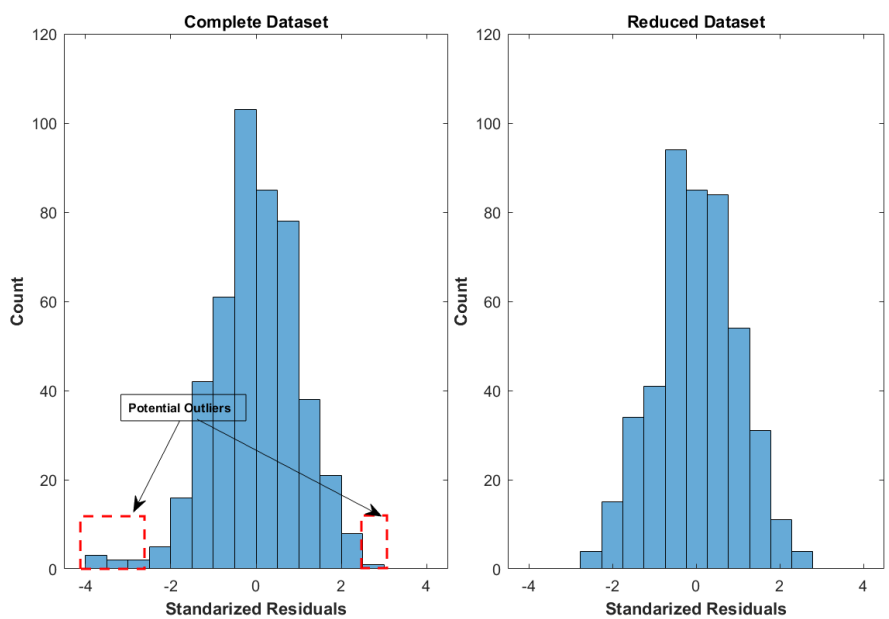


Figure S3: Comparison of residual histogram for the complete and reduced dataset after removal of outliers.

The reduced dataset shows that the outliers have been removed, and the distribution of residuals is more normal. The corresponding change can also be observed from the q-q plot shown below.

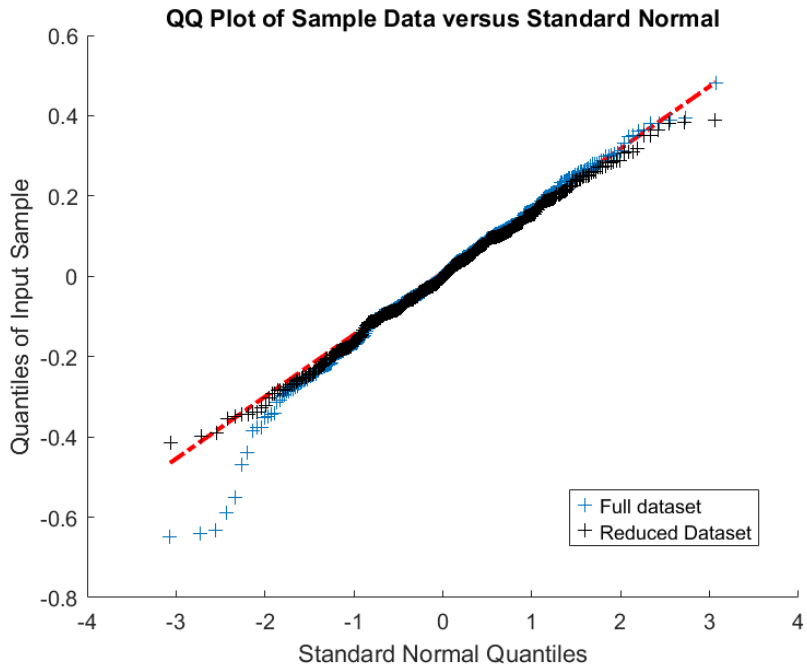


Figure S4: Comparison of residual q-q for the complete and reduced dataset after removal of outliers.

The details about the two models and the corresponding change in the fixed effect estimated have been tabulated below:

Table S3: Comparison of model statistics before and after dataset filtering for scaling parameter $W$

| Model | Number of data-points | p-values (SW test) | p-values (JB test) | Slope estimate | Intercept estimate |
|---|---|---|---|---|---|
| Complete dataset | 465 | $4.67 \times 10^{-4}$ | $1.00 \times 10^{-3}$ | -0.7704 | 0.313898 |
| Reduced dataset | 457 | 0.7343 | 0.3066 | -0.7731 | 0.313896 |

It can be observed that the p-values for the normality test increase, and the corresponding change in the fixed-effect estimates is minimal, i.e., $< 1\%$ for both the slope and the intercept. Similar results were obtained for other scaling parameters as well.